

Linear Algebra (Eng)

2026 S

Midterm

2026.4.19

Time Limit: 3 Hours

Name: _____

Student ID: _____

This exam includes 14 pages (including this page) and 5 problems. Please check to see if there is any missing page, and then write down your name and student ID number on this page and the first page of your answer sheets. Also write down the initials of your name on the top of every page of your answer sheets, in case they are scattered.

This exam allows one SINGLE cheat sheet (size A4, double-sided is fine). No calculator or any electronic device is allowed. Plagiarism of all kinds are strictly forbidden and will be severely punished.

Please write down your answers to the problems in the **SEPARATE ANSWER SHEETS**, and follow the following rules:

- **Always explain your answer.** You should always explain your answers. At least write down one sentence to explain your thoughts.
- **Write clearly and legibly.** Make sure that your writings can be read. The graders are NOT responsible to decipher illegible writings.
- **Partial credits will be given.**
- Blank spaces are provided in the exams. Feel free to use them as scratch papers. However, your formal answer has to be written in the **SEPARATE ANSWER SHEETS**, as required by the University.
- The total score of the exam is ???. If your total score exceeds ??? (there are ??? points in total), it will be recorded as ???.

Problems	Points	Score
1	11	
2	10	
3	14	
4	12	
5	15	
Total:	62	

1. (Matrix Recovery from RREF) Given a matrix $A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4]$, its Reduced Row Echelon Form is

$$R = \text{RREF}(A) = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Furthermore, we know:

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix}.$$

- (a) (3 points) Explicitly write out the full 3×4 matrix A with exact entries. (Reminder: You should NEVER just write an answer for any problems. Always show some process. Same for ALL problems for this entire exam.)
- (b) (3 points) Find a unit lower triangular matrix L such that $A = LR$.
- (c) (3 points) Find a basis for the Kernel of A .
- (d) (2 points) Determine the reduced row echelon form of the matrix $M = \begin{bmatrix} A & O \\ A & 2A \end{bmatrix}$.

Answer:

1. Note that R and A have the same column relations. Let $\mathbf{r}_j$ denote the j -th column of the RREF matrix R .

Observing the columns of R , it is evident that $\mathbf{r}_2 = 2\mathbf{r}_1$. So

$$\mathbf{a}_2 = 2\mathbf{a}_1 = 2 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix}.$$

Similarly, $\mathbf{r}_4 = -\mathbf{r}_1 + 3\mathbf{r}_3$. Therefore,

$$\mathbf{a}_4 = -\mathbf{a}_1 + 3\mathbf{a}_3 = -1 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \\ 12 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 13 \end{bmatrix}.$$

By assembling all the columns, the full, reconstructed matrix is explicitly written as:

$$A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 2 & 4 & 1 & 1 \\ -1 & -2 & 4 & 13 \end{bmatrix}.$$

2. Gaussian elimination gives

$$\begin{bmatrix} 1 & 2 & 0 & -1 \\ 2 & 4 & 1 & 1 \\ -1 & -2 & 4 & 13 \end{bmatrix} \mapsto \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 4 & 12 \end{bmatrix} \mapsto \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

By recording the row operations used in this process, we have

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 4 & 1 \end{bmatrix}.$$

Alternatively, note that $A\mathbf{e}_1 = L R \mathbf{e}_1 = L \mathbf{e}_1$, so the 1st column of A is the 1st column of L . Similarly, $A\mathbf{e}_3 = L R \mathbf{e}_3 = L \mathbf{e}_2$, so the 3rd column of A is the 2nd column of L . This also gives L directly.

3. By the Kernel-Finding algorithm in our lecture notes,

$$\text{RREF}(A) = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \mapsto \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \mapsto \begin{bmatrix} 0 & 2 & 0 & -1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

So a basis for kernel is made of $\begin{bmatrix} 2 \\ -1 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 0 \\ 3 \\ -1 \end{bmatrix}$. (For graders, we just need any two non-parallel vectors of the format $\begin{bmatrix} -2a+b \\ a \\ -3b \\ b \end{bmatrix}$.)

4. Block elimination gives

$$M = \begin{bmatrix} A & \\ A & 2A \end{bmatrix} \mapsto \begin{bmatrix} A & \\ & 2A \end{bmatrix} \mapsto \begin{bmatrix} A & \\ & A \end{bmatrix} \mapsto \begin{bmatrix} R & \\ & R \end{bmatrix}.$$

The last step is by applying $\begin{bmatrix} L^{-1} & \\ & L^{-1} \end{bmatrix}$. So we have

$$M \mapsto \begin{bmatrix} R & \\ & R \end{bmatrix} \mapsto \begin{bmatrix} 1 & 2 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \mapsto \begin{bmatrix} 1 & 2 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

2. (Geometry and Algebra of Skew-Symmetric Matrices) Let A be a 3×3 skew-symmetric matrix defined by

$$A = \begin{bmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{bmatrix}.$$

Let $\mathbf{a} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ be a vector in $\mathbb{R}^3$.

- (a) (4 points) We always have $A\mathbf{x} = \mathbf{a} \times \mathbf{x}$ for any $\mathbf{x} \in \mathbb{R}^3$, where $\times$ denotes the cross product. Using Gaussian elimination, find all solutions $\mathbf{x}$ to the equation $\mathbf{e}_1 \times \mathbf{x} = \mathbf{e}_2$.

- (b) (2 points) For what values of a, b, c is the matrix A invertible?

- (c) (2 points) Assume $c \neq 0$. Suppose for contradiction that A has an LU decomposition $A = LU$, where L is unit lower triangular and U is upper triangular. Show that the $(1, 1)$ entry of L or the $(1, 1)$ entry of U must be zero. Show that this leads to a contradiction.

- (d) (2 points) Prove that for any $n \times n$ skew-symmetric matrix M (where $M^T = -M$), its kernel is the orthogonal complement of its range.

Answer:

1. The equation $\mathbf{e}_1 \times \mathbf{x} = \mathbf{e}_2$ means $A\mathbf{x} = \mathbf{e}_2$ for $\mathbf{a} = \mathbf{e}_1$. Hence we have augmented matrix

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \mapsto \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \mapsto \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

So the solutions are $\mathbf{x} = \begin{bmatrix} k \\ 0 \\ -1 \end{bmatrix}$ for arbitrary k .

2. The matrix A is never invertible for any a, b, c . By the property of the cross product (or by direct calculation), $A\mathbf{a} = \mathbf{0}$. If $\mathbf{a} \neq \mathbf{0}$, then this means A is not invertible. If $\mathbf{a} = \mathbf{0}$, then A is the zero matrix, which is also not invertible.

3. Let $L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$ and $U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$. The $(1,1)$ entry of LU is $(l_{11})(u_{11}) + (0)(0) + (0)(0) = l_{11}u_{11}$. From the definition of A , the $(1,1)$ entry is 0. Therefore, $l_{11}u_{11} = 0$. So the $(1,1)$ entry of L or the $(1,1)$ entry of U must be zero.

If the $(1,1)$ entry of L is zero, then LU will have the first row entirely zero, hence $b = c = 0$, contradiction. If the $(1,1)$ entry of U is zero, then LU will have the first column entirely zero, hence $b = c = 0$, contradiction.

4. From the Fundamental Theorem of Linear Algebra, we know that

$$\text{Ker}(A)^\perp = \text{Ran}(A^T) = \text{Ran}(-A) = \text{Ran}(A).$$

3. (Block Matrix Properties and Decompositions) Let M be a block matrix below

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

Here B is invertible and 2×2 , and C is invertible and 3×3 .

- (a) (3 points) Which of the following computations are defined? $D - CA^{-1}B$, $C - AB^{-1}D$, $B - AC^{-1}D$. (For the one not defined, remember to tell me why. Recall that for all subproblems you should give process!)

- (b) (3 points) Suppose we have the following equation for vectors $\mathbf{x}, \mathbf{w} \in \mathbb{R}^3$ and $\mathbf{y}, \mathbf{v} \in \mathbb{R}^2$:

$$M \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{v} \\ \mathbf{w} \end{bmatrix}.$$

Fix an arbitrary 3×2 matrix P and an arbitrary 3×3 matrix Q . Find matrix S expressed in terms of A, B, C, D, P, Q , such that $S \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = P\mathbf{v} + Q\mathbf{w}$.

- (c) (4 points) Find the missing block denoted by “??” and the elementary block shearing matrices E_1 and E_2 , such that we have the following decomposition:

$$M = E_1 \begin{bmatrix} O & ?? \\ C & O \end{bmatrix} E_2.$$

- (d) (4 points) Suppose A is entirely zero in our block matrix. Find M^{-1} in terms of B, C , and D .

Answer:

1. Analyzing the options:

Illegal. A is 2×3 and therefore not square, so A^{-1} does not exist.

Illegal. A is 2×3 and B^{-1} is 2×2 . The inner dimensions (3 and 2) do not match, so AB^{-1} cannot be computed.

Legal. The product $AC^{-1}D$ has dimensions $(2 \times 3) \times (3 \times 3) \times (3 \times 2) \implies 2 \times 2$. Since B is also 2×2 , the subtraction is well-defined.

2. Expanding the given block matrix equation yields a system of two equations:

$$Ax + By = v,$$

$$Cx + Dy = w.$$

So we have

$$Pv + Qw = PAx + PBy + QCx + QDy = [PA + QC \quad PB + QD] \begin{bmatrix} x \\ y \end{bmatrix}.$$

$$\text{So } S = [PA + QC \quad PB + QD].$$

3. We use C to kill the entry above with a row operation, and to kill the entry to the right of C by a column operation. We have

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} I & AC^{-1} \\ & I \end{bmatrix} \begin{bmatrix} C & B - AC^{-1}D \\ & D \end{bmatrix} = \begin{bmatrix} I & AC^{-1} \\ & I \end{bmatrix} \begin{bmatrix} C & B - AC^{-1}D \\ & D \end{bmatrix} \begin{bmatrix} I & C^{-1}D \\ & I \end{bmatrix}.$$

$$\text{So the missing block is } B - AC^{-1}D \text{ and } E_1 = \begin{bmatrix} I & AC^{-1} \\ & I \end{bmatrix} \text{ and } E_2 = \begin{bmatrix} I & C^{-1}D \\ & I \end{bmatrix}.$$

4. With $A = O$, $M = \begin{bmatrix} O & B \\ C & D \end{bmatrix}$. To find the inverse, we have

$$\begin{bmatrix} B & I \\ C & D \end{bmatrix} \mapsto \begin{bmatrix} C & D & I \\ & B & I \end{bmatrix} \mapsto \begin{bmatrix} I & C^{-1}D & C^{-1} \\ & I & B^{-1} \end{bmatrix} \mapsto \begin{bmatrix} I & -C^{-1}DB^{-1} & C^{-1} \\ & I & B^{-1} \end{bmatrix}.$$

$$\text{So the answer is } \begin{bmatrix} O & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} -C^{-1}DB^{-1} & C^{-1} \\ & B^{-1} \end{bmatrix}$$

4. (Inclusion-Exclusion Principle and Projections)

Consider the following two subspaces inside $\mathbb{R}^4$

$$V = \text{span} \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 : x_1 - 2x_2 + x_3 = 0 \text{ and } x_4 = 0 \right\}.$$

$$W = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 0 \right\}.$$

- (a) (4 points) Use rank-nullity theorem to find the dimension of V and W , and then find basis for V and W .
- (b) (2 points) Find the dimension of $V + W$ and $V \cap W$.
- (c) (4 points) Find the 4×4 orthogonal projection matrix P onto the space $V \cap W$, and the 4×4 orthogonal projection matrix Q onto the orthogonal complement of $V \cap W$.
- (d) (2 points) Compute the explicit value of $(I - 2P)^{-1} + (I - 2Q)^3$.

Answer:

1. We know $V = \text{Ker}(A)$ where $A = \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$. The matrix A is already in row-echelon form and has rank 2. By the Rank-Nullity Theorem, $\dim(V) = 4 - \text{rank}(A) = 4 - 2 = 2$. A basis for V could be $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$. But any

two non-parallel vectors of the format $\begin{bmatrix} a \\ a+b \\ b \\ 0 \end{bmatrix}$ will do.

Similarly, $W = \text{Ker}(B)$ where $B = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$. Here $\text{rank}(B) = 1$, so $\dim(W) = 4 - 1 = 3$. A basis for W could be $\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$. But

any three linearly independent vectors of the format $\begin{bmatrix} a \\ b \\ c \\ -a-b-c \end{bmatrix}$ will do.

2. Note that a vector $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$ in V is outside of W , so $\dim(V + W) > \dim(W) = 3$.

But everything is in $\mathbb{R}^4$. Hence $\dim(V + W) = 4$.

By inclusion exclusion principle, $\dim(V \cap W) = \dim V + \dim W - \dim(V + W) = 2 + 3 - 4 = 1$.

3. The subspace $V \cap W$ is spanned by $\mathbf{u} = \begin{bmatrix} 1 & 0 & -1 & 0 \end{bmatrix}^T$. The orthogonal projection matrix P is:

$$P = \frac{\mathbf{u}\mathbf{u}^T}{\mathbf{u}^T\mathbf{u}} = \frac{1}{2} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The projection Q onto the orthogonal complement $(V \cap W)^\perp$ is:

$$Q = I - P = \frac{1}{2} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

4. Note that $I - 2P, I - 2Q$ are reflections. Hence $(I - 2P)^{-1} = I - 2P$ and $(I - 2Q)^3 = I - 2Q$. So we have

$$(I - 2P)^{-1} + (I - 2Q)^3 = (I - 2P) + (I - 2Q) = 2I - 2(P + Q) = 2I - 2I = O.$$

Here note that $P + Q = I$ as they are complementary projections. The result is the 4×4 all zero matrix.

5. (Polynomial Function Spaces) Let V be the abstract vector space of all polynomials of degree at most 2 with real coefficients. Consider the transformation $L : V \rightarrow V$ defined as:

$$L(p(x)) = xp'(x) + p(x+1).$$

where $p'(x)$ denotes the derivative of $p(x)$.

- (a) (4 points) Let D be the operator sending $p(x)$ to $p'(x)$. Let M be the operator sending $p(x)$ to $xp(x)$. Let S be the operator sending $p(x)$ to $p(x+1)$. Show that S is linear by definition of a linear map, and then write L in terms of D, M, S (hence L is linear).

- (b) (4 points) Find the matrix representation of the linear map L relative to the basis $\mathcal{B} = \{1, x, x^2\}$ for both the domain and the codomain.

- (c) (4 points) Consider an alternative basis $\mathcal{C} = \{1, x-1, (x+1)^2\}$. Find the change-of-coordinate matrix $I_{\mathcal{C} \leftarrow \mathcal{B}}$ and $I_{\mathcal{B} \leftarrow \mathcal{C}}$. (Please clearly indicate in your answer which is which! If you just throw two matrices at me then I have no idea which one is which process.)

- (d) (3 points) Utilizing the matrices derived in the previous sections, determine the matrix representation $L_{\mathcal{C} \leftarrow \mathcal{C}}$.

Answer:

1. To show S is linear, we verify the definition for any polynomials $p(x), q(x) \in V$ and scalar $c \in \mathbb{R}$:

Additivity: $S((p+q)(x)) = (p+q)(x+1) = p(x+1) + q(x+1) = S(p(x)) + S(q(x))$.

Homogeneity: $S((cp)(x)) = (cp)(x+1) = cp(x+1) = cS(p(x))$.

The operator L can be expressed as:

$$L = MD + S.$$

Since M, D , and S are linear operators on V , and the set of linear operators is closed under addition and composition, L is necessarily linear.

2. We compute the image of each basis element in $\mathcal{B} = \{1, x, x^2\}$ under L :

$$L(1) = x(0) + 1 = 1.$$

$$L(x) = x(1) + (x+1) = 2x + 1.$$

$$L(x^2) = x(2x) + (x+1)^2 = 2x^2 + (x^2 + 2x + 1) = 3x^2 + 2x + 1.$$

Placing the coordinates of these images into the columns, the matrix representation is:

$$L_{\mathcal{B} \leftarrow \mathcal{B}} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}.$$

3. The matrix $I_{\mathcal{B} \leftarrow \mathcal{C}}$ is easier. We simply express $\mathcal{C} = \{1, x-1, (x+1)^2\}$ in terms of $\mathcal{B}$. Clearly $(x+1)^2 = x^2 + 2x + 1$, and the other two is immediate. So we have

$$I_{\mathcal{B} \leftarrow \mathcal{C}} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

We find $I_{\mathcal{C} \leftarrow \mathcal{B}}$ by inverse, and get

$$I_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} 1 & 1 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

4. We compute $L_{\mathcal{C} \leftarrow \mathcal{C}} = I_{\mathcal{C} \leftarrow \mathcal{B}} L_{\mathcal{B} \leftarrow \mathcal{B}} I_{\mathcal{B} \leftarrow \mathcal{C}}$:

$$L_{\mathcal{C} \leftarrow \mathcal{C}} = \begin{bmatrix} 1 & 1 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

This can be computed as (multiplying the first two matrices first)

$$L_{\mathcal{C} \leftarrow \mathcal{C}} = \begin{bmatrix} 1 & 3 & -6 \\ 0 & 2 & -4 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Or as (multiplying the last two matrices first)

$$L_{\mathcal{C} \leftarrow \mathcal{C}} = \begin{bmatrix} 1 & 1 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$